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1. Bessel Process

A common object of study is the Bessel process, which is the norm of an n-dimensional Brownian motion. To study its law, we may naïvely apply Itô's lemma to the function $f(x) = \|x\|$. Letting $R_t = \|B_t\|$, we have:

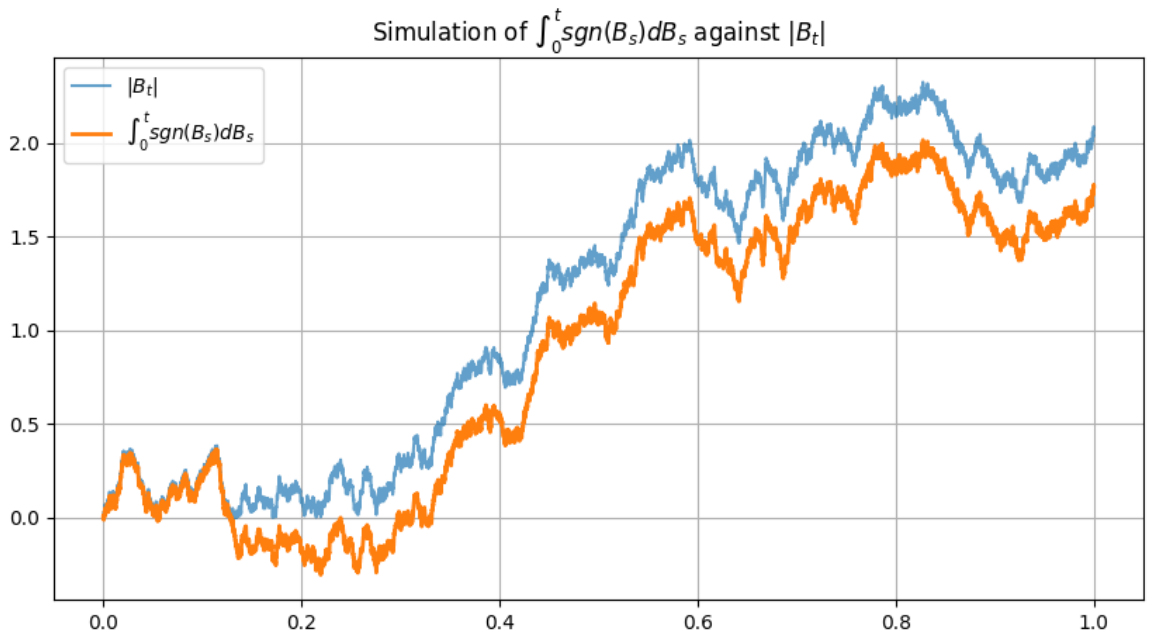
$$\mathrm{d}R_t = \frac{B_t \cdot \mathrm{d}B_t}{R_t} + \frac{n-1}{R_t}\mathrm{d}t$$

This is not immediately justified, since f is not twice differentiable at the origin, however, it's not hard to show, with a mollification argument, that this is indeed correct.

So let's naïvely apply this to $n = 1$. Then we have:

$$|B_t| = \int_0^t \mathrm{sgn}(B_s)\mathrm{d}B_s$$

But this is wrong! We will see how this is wrong, but a quick simulation shows that the two sides are not equal:



To solve this, we introduce the concept of local time. Intuitively, local time is a stochastic process that measures how much time another stochastic process spends at a particular point. We consider Brownian motion, and are interested in the quantity

$$L_t(x) = \lim_{\epsilon \downarrow 0} \frac{1}{2\epsilon} \int_0^t [B_s \in (x-\epsilon, x+\epsilon)]\mathrm{d}s$$

Which, if the integrand were continuous, this would exist. The indicator function is not very regular, so we look at $f \in C_2(\mathbb{R})$, and investigate

$$\frac{1}{2}\int_0^t f(B_s)\mathrm{d}s = \int_{\mathbb{R}} f(x)L_t(x)\mathrm{d}x$$

If we stare at Itô's lemma for a while, we see that the second derivative term is suspiciously similar to the left hand side of the above integral, if we have some function F such that $F'' = f$. A neat trick is to let

$$F(x) = \int_{\mathbb{R}} f(y)(x-y)^+\mathrm{d}y$$

(which is also a portfolio of call options with strike y). By Itô's lemma, we have

$$\begin{aligned}
\mathrm{d}F(B_t) &= \int_{-\infty}^{B_t} f(y)\mathrm{d}y\,\mathrm{d}B_t + \frac{1}{2}f(B_t)\mathrm{d}t \\
F(B_t) &= F(B_0) + \int_0^t \int_{\mathbb{R}} f(y)[B_s > y]\mathrm{d}y\,\mathrm{d}B_s + \frac{1}{2}\int_0^t f(B_s)\mathrm{d}s \\
&= F(B_0) + \int_{\mathbb{R}} f(y) \int_0^t [B_s > y]\mathrm{d}B_s\,\mathrm{d}y + \frac{1}{2}\int_0^t f(B_s)\mathrm{d}s \\
\implies \int_0^t f(B_s)\mathrm{d}s &= 2\int_{\mathbb{R}} f(y)\left((B_t - y)^+ - (B_0 - y)^+ - \int_0^t [B_s > y]\mathrm{d}B_s\right) dy
\end{aligned}$$

Which follows if you believe in a stochastic version of Fubini's theorem. This motivates Tanaka's formula:

$$L_t(x) = (B_t - x)^+ - (B_0 - x)^+ - \int_0^t [B_s > x]\mathrm{d}B_s$$

So now, how does this help us? The following argument can be completed as an exercise, but considering the mollification of $f(x) = |x|$ with

$$g_\epsilon(x) = \begin{cases} \frac{x^2}{2\epsilon} + \frac{\epsilon}{2} & |x| \le \epsilon \\ |x| & |x| > \epsilon \end{cases}$$

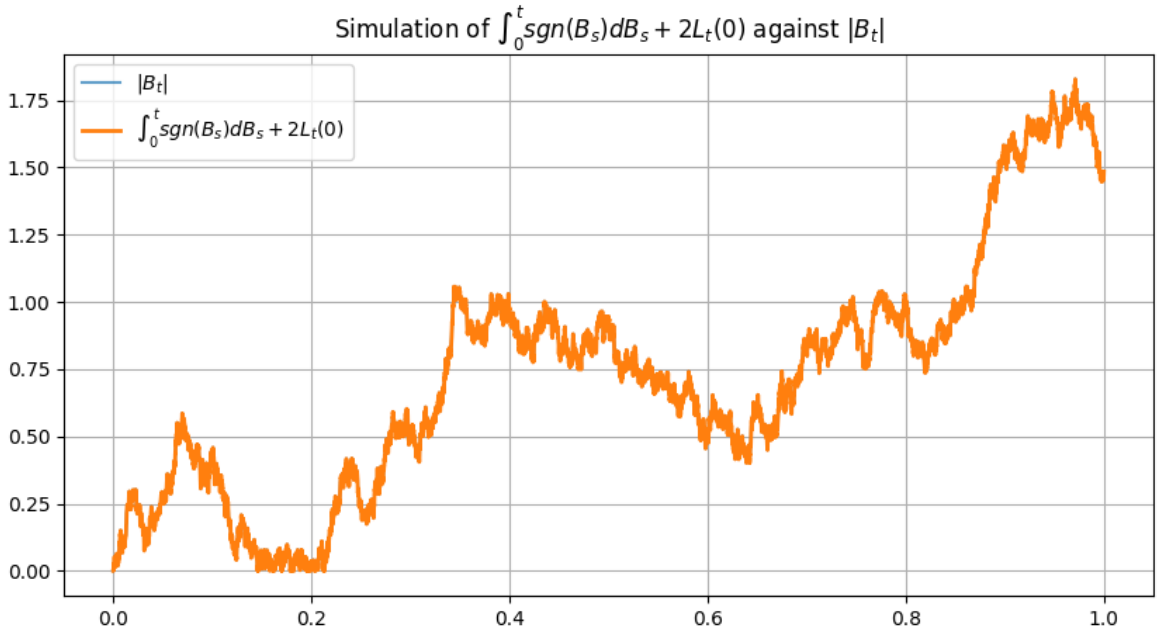
We can use Itô's lemma on g_ϵ and apply our local time result to obtain the following:

$$\begin{aligned}
g_\epsilon(B_t) - g_\epsilon(B_0) &= \int_0^t g_\epsilon'(B_s)\mathrm{d}B_s + \frac{1}{2}\int_0^t g_\epsilon''(B_s)\mathrm{d}s \\
&= \int_0^t g_\epsilon'(B_s)\mathrm{d}B_s + \int_{\mathbb{R}} g_\epsilon''(x)L_t(x)\mathrm{d}x \\
&= \int_0^t g_\epsilon'(B_s)\mathrm{d}B_s + 2\int_{-\epsilon}^{\epsilon} \frac{1}{2\epsilon}L_t(x)\mathrm{d}x
\end{aligned}$$

Taking limits, we obtain Tanaka's formula for absolute value:

$$|B_t| = \int_0^t \mathrm{sgn}(B_s)\mathrm{d}B_s + 2L_t(0)$$

A quick simulation shows that this is indeed correct:



We've covered the first half of the title, now onto the second half: hedging. What does any of this have to do with hedging?

2. Hedging

This section is inspired by George Lowther's blog almost sure math.

Let's consider a European call option with strike K and expiry T, and lets assume this call is out of the money, i.e. $S_0 < K$. A hedge is a trading strategy on the underlying asset that replicates the payoff of the option at expiry.

A naïve hedge for a European call option is to hold one unit of the underlying asset if the asset price is at least K, and hold nothing otherwise. By this strategy, the value of the porfolio at time T is $(S_T - K)^+$, which is exactly the payoff of the call option.

An immediate probelm with this strategy is that it presents an arbitrage opportunity! We buy for K, and if it drops below K, we lose no money, and if the stock stays above K we make $(S_T - K)^+$. The issue is that we are requiring the trader to make uncountably many trades. More precisely, the stock price process has unbounded variation, so if X_t is our PnL process, we do not have $X_T = (S_T - K)^+$. However, we can fix this using local time. The PnL of the trading strategy is given by:

$$X_T = \int_0^T [S_t > K] \mathrm{d}S_t$$

For simplicity, (and WLOG by Girsanov transformation), S is a Brownian motion. Then we immediately recognize our integral!

$$X_T = (S_T - K)^+ - L_T(K)$$

We no longer have replication, but we no longer have arbitrage either. Additionally, we can compute the law of our PnL. Assuming $T = 1$ and $K = 0$, as George does in his blog post, letting φ be the standard normal density, we have:

$$p_{X_1}(x) = \begin{cases} \frac{2}{3}\varphi(x) & x > 0 \\ \frac{8}{3}\varphi(2x) & x < 0 \end{cases}$$